

A charge-pump PLL on Tiny Tapeout GF180

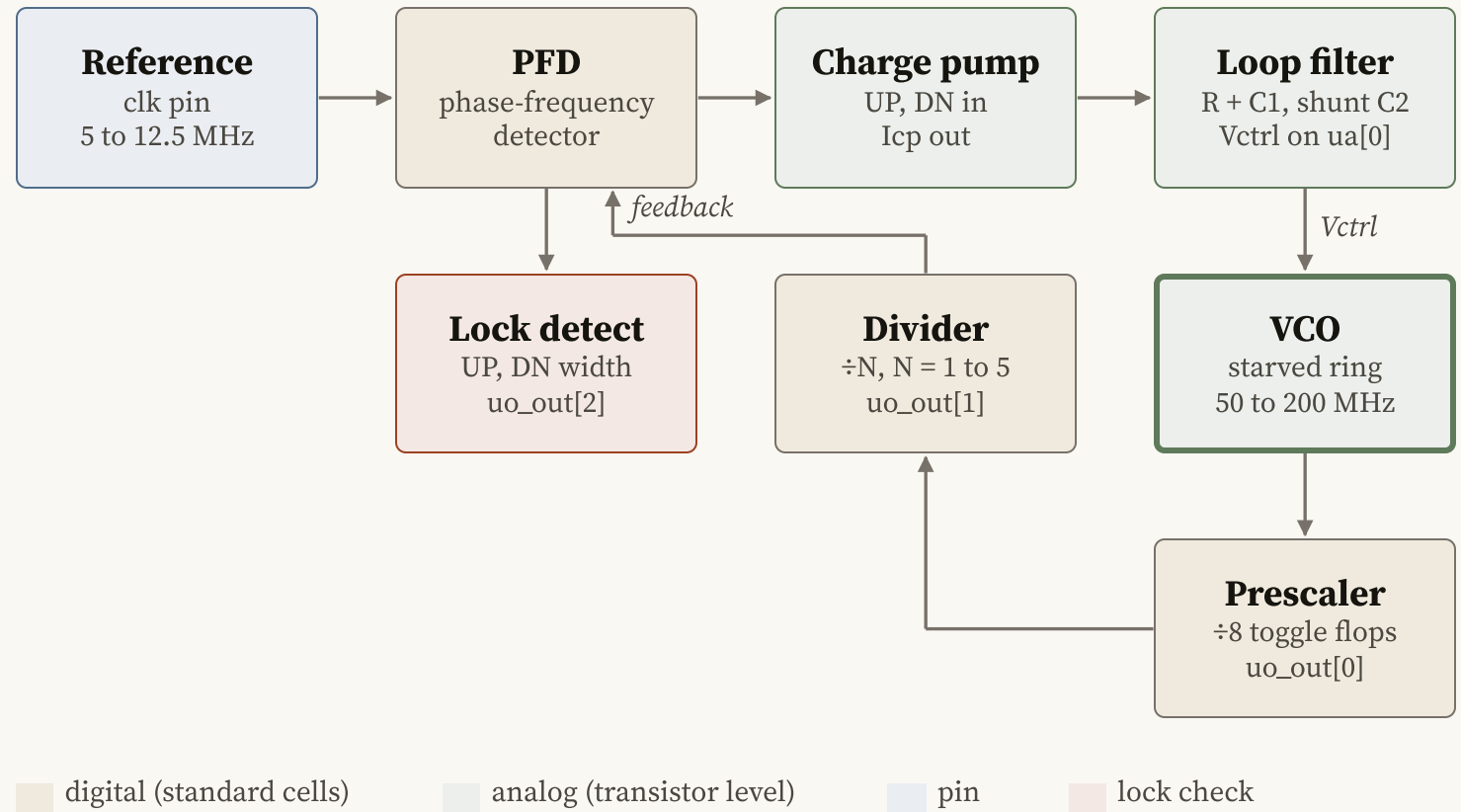
Statement of purpose

- Build a charge-pump PLL: the clock multiplier every chip uses to make a fast clock from a slow one.
- Multiply a 5 to 12.5 MHz reference by $8N$ ($N = 1$ to 5) onto a 50 to 200 MHz on-chip VCO.
- Mixed-signal: a standard-cell PFD and dividers around a transistor-level pump, filter and VCO.

Testing the whole loop

- Linear model (python-control, ngspice AC): margin, bandwidth, lock time
- Real RTL in cocotb with a Python pump, filter and VCO: locks at every N
- Co-simulation (cocotbext-ams): RTL with the transistor pump and filter
- Transistor loop in ngspice from near lock, plus one overnight lock run

System diagram



Pin assignment and proposed specification

IO pin assignment (Tiny Tapeout analog tile)

Pin	Dir	Use
clk	in	reference, 5 to 12.5 MHz
rst_n, ena	in	standard reset and enable
ui_in[2:0]	in	N, the divide ratio (1 to 5)
ui_in[5:3]	in	spare
ui_in[6]	in	PLL enable (powers down pump, VCO)
ui_in[7]	in	observe VCO $\div 8$ or $\div 16$
uo_out[0]	out	VCO $\div 8$ or $\div 16$
uo_out[1]	out	feedback divider output
uo_out[2]	out	lock
uo_out[4:3]	out	UP, DN (debug)
uo_out[7:5]	out	spare, driven low
uio[1:0]	in	charge-pump current trim
uio[7:2]	in	spare
ua[0]	analog	Vctrl, the loop-filter node
ua[1]	analog	bias reference (external resistor)

Proposed specification

Parameter	Target
Reference	5 to 12.5 MHz (nominal 6.25 MHz)
Multiplication	8N with N = 1 to 5, so 8 $\times$ to 40 $\times$
Nominal output	6.25 MHz $\times$ 16 = 100 MHz
VCO range	50 to 200 MHz, typical corner
Prescaler	fixed $\div 8$, so all logic runs $\leq$ 50 MHz
Charge pump	I _{cp} $\approx$ 20 μ A, 2-bit trim
Loop filter	R $\approx$ 26 k Ω , C1 $\approx$ 20 pF, C2 $\approx$ C1/10
Phase margin	$\geq 45^\circ$ at N = 2 (8N = 16)
Lock time	under 50 μ s
Observed output	VCO $\div 8$ or $\div 16$ on uo_out[0]

First ngspice runs at the typical corner: I_{cp} 19 μ A, VCO 10 to 258 MHz, phase margin 49.5 $^\circ$ at 8N = 16.

Timeline for completion and who does what

